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Abstract will come here...

I. INTRODUCTION

Wet-node boundary conditions, where the boundary is defined to be infinitesimally close to the fluid nodes, have been shown in various test cases to be able to achieve higher accuracy at straight boundaries than link-wise boundary conditions, where the boundary is defined to be half a grid-point away from the fluid nodes. This has been shown for velocity boundaries [1] and pressure boundaries [2]. Especially methods that conserve the known particle populations, which stream in from the bulk of the domain, are very accurate at moderate Reynolds number [1]. For instance, the Non-Equilibrium Bounce-Back (NEBB) method [2] has been shown to be third-order accurate for prescribing the no-slip condition on straight boundaries for single-component fluids in the absence of external forcing [3].

In this paper, we aim to expand upon the NEBB method to enable its use in more complex flow problems with straight boundaries and moderate Reynolds numbers. For this, we define a stencil-independent formulation of the force-corrected and mass-corrected multi-component NEBB method.

Stencil-independent formulations for single-component fluids have been posed in previous research [3][4], but not in the presence of external forces. Force-corrections have been introduced in previous research too, in the context of single-component pseudopotential wetting simulations at a no-slip boundary [5], and two-component pseudopotential simulations at an outlet [6]. However, both of these are formulated specifically for the D2Q9.

Furthermore, mass corrections have also been defined in previous research for single-component fluids in the absence of external forces [4]. Finally, a multi-component formulation of the NEBB has been derived two-component pseudopotential simulations [6], but again specifically for the D2Q9. This method also does not take mass corrections into account and simplifications made in the derivation can lead to deviations of the density and velocity from the imposed values.

After the derivation of the expanded NEBB method, we will use an asymptotic analysis in Knudsen number [7] to assess its accuracy and the validity of the mass correction. Next, we will perform various numerical test cases for the expanded NEBB against the widely-used Halfway Bouce-Back (HWBB) method, both for single-component fluids and immiscible multi-component fluids. The multi-component simulations will be performedd with both the Shan-Chen method [8] and the Color-Gradient method [9].

II. MULTI-COMPONENT LATTICE BOLTZMANN METHOD

The Lattice Boltzmann Equation using the BGK collision operator [10] for a two-component system is given as follows,

$$f_i^\sigma(\boldsymbol{x}+\boldsymbol{c}_i, t+1) - f_i^\sigma = -\frac{1}{\tau^\sigma}\left(f_i^\sigma - f_i^{\mathrm{eq},\sigma}\right), \tag{1}$$

where σ indicates the fluid component and τ^σ gives the corresponding relaxation time. As is common for the Lattice Boltzmann method, the time step and grid spacing are set to unity. Lastly, the discrete velocity directions are indicated by $\boldsymbol{c}_i$. The macroscopic quantities can be derived from the distribution functions,

$$\rho^\sigma = \sum_i f_i^\sigma, \quad \rho\boldsymbol{u} = \sum_\sigma\sum_i f_i^\sigma \boldsymbol{c}_i, \tag{2}$$

where $\rho = \sum_\sigma \rho^\sigma$. Two different methods will be used separate the two fluids and create surface tension. The Shan-Chen method uses a repulsion force between the two fluids [8] and the Color-Gradient method adds surface tension and repulsion using two extra collision operators [9].

A. The Shan-Chen Method

The Shan-Chen force is given as follows [8],

$$\boldsymbol{F}_{\mathrm{SC}}^\sigma(\boldsymbol{x}) = -G\rho^\sigma(\boldsymbol{x})\sum_i w_i \rho^{\tilde{\sigma}}\left(\boldsymbol{x}+\boldsymbol{c}_i\right)\boldsymbol{c}_i, \tag{3}$$

where $\tilde{\sigma}$ indicates the other fluid component and G is the interaction strength. Furthermore, w_i indicates the weight attributed to the velocity component $\boldsymbol{c}_i$. A source term S_i^σ is introduced on the right-hand-side of Eq. (1) to incorporate this force [11],

$$S_i^\sigma = \left(1 - \frac{1}{2\tau^\sigma}\right) w_i \left(\frac{\boldsymbol{c}_i - \boldsymbol{u}}{c_s^2} + \frac{\boldsymbol{c}_i\cdot\boldsymbol{u}}{c_s^4}\boldsymbol{c}_i\right)\cdot\boldsymbol{F}^\sigma, \tag{4}$$

where $c_s = \sqrt{1/3}$. Furthermore, the velocity given in Eq. (2) is shifted by the external forcing,

$$\rho\boldsymbol{u} = \sum_\sigma\sum_i f_i^\sigma\boldsymbol{c}_i + \frac{1}{2}\sum_\sigma \boldsymbol{F}^\sigma. \tag{5}$$

Finally, the equilibrium distribution is computed as follows,

$$f_i^{\mathrm{eq},\sigma} = w_i\rho^\sigma\left(1 + \frac{\boldsymbol{u}\cdot\boldsymbol{c}_i}{c_s^2} + \frac{(\boldsymbol{u}\cdot\boldsymbol{c}_i)^2}{2c_s^4} - \frac{|\boldsymbol{u}|^2}{2c_s^2}\right). \tag{6}$$

B. The Color-Gradient Method

The Lattice Boltzmann Equation is usually split in a collision and streaming step,

$$\begin{aligned} f_i^{*,\sigma}(\boldsymbol{x},t) &= f_i^{\sigma}(\boldsymbol{x},t) - \frac{1}{\tau^{\sigma}}\left(f_i^{\sigma} - f_i^{\mathrm{eq},\sigma}\right), \\ f_i^{\sigma}\left(\boldsymbol{x},t+1\right) &= f_i^{*,\sigma}\left(\boldsymbol{x}-\boldsymbol{c}_i,t\right). \end{aligned} \tag{7}$$

The Color-Gradient method introduces two additional steps after the collision step to introduce interface tension and phase separation. The equilibrium distribution is adapted to support separate speeds of sound in the fluids [9],

$$f_i^{\mathrm{eq},\sigma} = \rho_w^{\sigma}\left[\phi_i^{\sigma} + w_i\left(\frac{\boldsymbol{u}\cdot\boldsymbol{c}_i}{c_s^2} + \frac{\left(\boldsymbol{u}\cdot\boldsymbol{c}_i\right)^2}{2c_s^4} - \frac{|\boldsymbol{u}|^2}{2c_s^2}\right)\right] + \Phi_i^{\sigma}, \tag{8}$$

where the values for ϕ_i^{σ} are given in Tab. (I).

$\lvert\boldsymbol{c}_i\rvert^2$	0	1	2	3
D2Q9	α^{σ}	$(1-\alpha^{\sigma})/5$	$(1-\alpha^{\sigma})/20$	-
D3Q19	α^{σ}	$(1-\alpha^{\sigma})/12$	$(1-\alpha^{\sigma})/24$	-
D3Q27	α^{σ}	$2(1-\alpha^{\sigma})/19$	$(1-\alpha^{\sigma})/38$	$(1-\alpha^{\sigma})/152$

TABLE I. Values for ϕ_i^{σ} for different lattices [9].

$\lvert\boldsymbol{c}_i\rvert^2$	0	1	2	3
D2Q9	$-8/3$	$-1/6$	$1/12$	-
D3Q19	$-5/2$	$-1/6$	$1/24$	-
D3Q27	$-20/9$	$-2/9$	$1/36$	$1/36$

TABLE II. Values for ψ_i for different lattices [9].

$\lvert\boldsymbol{c}_i\rvert^2$	0	1	2	3
D2Q9	0	1/2	1/8	-
D3Q19	0	1/4	1/8	-
D3Q27	0	1/3	1/12	1/48

TABLE III. Values for ξ_i for different lattices [9].

The values for α^{σ} depend on the initial density ratio between the fluids,

$$\gamma = \frac{\rho_0^{\sigma}}{\rho_0^{\tilde{\sigma}}} = \frac{1-\alpha^{\tilde{\sigma}}}{1-\alpha^{\sigma}}. \tag{9}$$

We choose $\alpha^{\tilde{\sigma}} = 0.2$ [12], where $\tilde{\sigma}$ corresponds to the low-density component. This gives $\alpha^{\sigma} = 1 - \left(1-\alpha^{\tilde{\sigma}}\right)/\gamma$. The pressure is now computed as follows,

$$p = \sum_{\sigma} p^{\sigma} = \sum_{\sigma} \rho_w^{\sigma}\zeta\left(1-\alpha^{\sigma}\right), \tag{10}$$

where ζ is a parameter depending on the lattice. For the D2Q9, D3Q19 and D3Q27, we have $\zeta = 3/5$, $\zeta = 1/2$ and $\zeta = 9/19$, respectively [9]. Finally, the Equilibrium enhancement Φ_i^{σ} is defined as follows,

$$\Phi_i^{\sigma} = \bar{\nu}\left[\psi_i u_{\alpha}\partial_{\alpha}\rho^{\sigma} + \xi_i c_{i\alpha}c_{i\beta}\left(u_{\alpha}\partial_{\beta}\rho^{\sigma} + u_{\beta}\partial_{\alpha}\rho^{\sigma}\right)\right], \tag{11}$$

where we sum over repeated Greek indices. The stencil-dependent constants ψ_i and ξ_i are given in Tabs. (II) and (III), respectively.

We will also need the effective relaxation parameter on the interface. This is computed as follows [9],

$$\begin{aligned} \frac{1}{\bar{\nu}} &= \frac{\rho^{\sigma}}{\rho}\frac{1}{\nu^{\sigma}} + \frac{\rho^{\tilde{\sigma}}}{\rho}\frac{1}{\nu^{\tilde{\sigma}}}, \\ \omega^{\mathrm{eff}} &= \frac{2}{6\bar{\nu}+1}. \end{aligned} \tag{12}$$

Note that this effective collision frequency ω^{eff} will also be used in place of $1/\tau^{\sigma}$ in the BGK collision operator for both components [13].

The distribution after the original collision step in Eq. (7) is relabeled as $f_i^{\mathrm{coll},\sigma}$. Next, we perform the perturbation step [9],

$$f_i^{\mathrm{pert},\sigma} = f_i^{\mathrm{coll},\sigma} + A|\boldsymbol{G}|\left(w_i\frac{\left(\boldsymbol{G}\cdot\boldsymbol{c}_i\right)^2}{|\boldsymbol{G}|^2} - B_i\right), \tag{13}$$

where $A = \frac{9}{8}\omega^{\mathrm{eff}}\sigma$, with σ as the surface tension. The factors B_i are stencil dependent and are given in Tab. (IV).

$\lvert\boldsymbol{c}_i\rvert^2$	0	1	2	3
D2Q9	-4/27	2/27	5/108	-
D3Q19	-2/9	1/54	1/27	-
D3Q27	-16/81	2/81	2/81	7/648

TABLE IV. Values for B_i for different lattices [9].

Lastly, we introduce the color gradient $\boldsymbol{G}$ [9],

$$\boldsymbol{G} = \boldsymbol{\nabla}\rho^N \approx \frac{1}{c_s^2}\sum_i w_i\rho^N(\boldsymbol{x}+\boldsymbol{c}_i)\boldsymbol{c}_i, \tag{14}$$

with $\rho^N = \left(\rho^{\sigma} - \rho^{\tilde{\sigma}}\right)/\rho$. Note that the density gradients in the equilibrium distribution are computed in the same way. Finally, we will perform the recoloring step [14],

$$\begin{aligned} f_i^{*,\sigma} &= \frac{\rho^{\sigma}}{\rho}\sum_{\sigma} f_i^{\mathrm{pert},\sigma} + \beta\frac{\rho^{\sigma}\rho^{\tilde{\sigma}}}{\rho^2}\frac{\boldsymbol{G}\cdot\boldsymbol{c}_i}{|\boldsymbol{G}||\boldsymbol{c}_i|}\sum_{\sigma} f_i^{\mathrm{eq},\sigma}|_{\boldsymbol{u}=\boldsymbol{0}}, \\ f_i^{*,\tilde{\sigma}} &= \frac{\rho^{\tilde{\sigma}}}{\rho}\sum_{\sigma} f_i^{\mathrm{pert},\sigma} - \beta\frac{\rho^{\sigma}\rho^{\tilde{\sigma}}}{\rho^2}\frac{\boldsymbol{G}\cdot\boldsymbol{c}_i}{|\boldsymbol{G}||\boldsymbol{c}_i|}\sum_{\sigma} f_i^{\mathrm{eq},\sigma}|_{\boldsymbol{u}=\boldsymbol{0}}, \end{aligned} \tag{15}$$

where $\tilde{\sigma}$ is again the low density component. Additionally, the parameter β tunes the interface thickness, for which $\beta = 0.7$ [12] is chosen.

III. MULTI-COMPONENT BOUNDARY CONDITION

The Non-Equilibrium Bounce-Back scheme [2] is a boundary condition based on the wet-node approach. Instead of the more common Half-way Bounce-Back scheme where the boundary fluid nodes are half a node away from the physical wall, the boundary fluid nodes are infinitesimally close to the physical wall in the wet-node approach. The NEBB scheme leaves the populations streaming out of the domain unchanged after streaming. The unknown populations streaming into the domain can be found using the non-equilibrium parts of the populations streaming out of the domain. Additionally, momentum corrections N_α, where α is the spatial component, are introduced [3]. This gives the following boundary condition for the unknown incoming populations i,

$$\begin{aligned} f_i^{\mathrm{neq},\sigma} &= f_q^{\mathrm{neq},\sigma} - w_i c_{i\alpha} N_\alpha^\sigma, \\ f_i^\sigma &= f_q^\sigma + 2 w_i \rho^\sigma \frac{\boldsymbol{u}\cdot\boldsymbol{c}_i}{c_s^2} - w_i c_{i\alpha} N_\alpha^\sigma. \end{aligned} \tag{16}$$

where the index q is given by $\boldsymbol{c}_i = -\boldsymbol{c}_q$. Furthermore, repeated Greek indices indicate sums. Note that the incoming populations are defined by $\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}}_w > 0$, where $\hat{\boldsymbol{n}}_w$ is the normal vector to the boundary pointing into the domain. The density ρ and the velocity $\boldsymbol{u}$ are the imposed boundary density and velocity. To conserve mass, we will satisfy the condition that the amount of mass being removed from the domain is equal to the mass added to the domain at every grid point. During the streaming step, the values of the outgoing populations f_q will be overwritten by populations from the bulk, leading to mass loss. The mass gain is given by Eq. (16). This gives the net mass loss at the boundary due to the NEBB method,

$$\begin{aligned} \Delta m^\sigma &= \sum_i^{\mathrm{out}} \left(f_i^{*,\sigma} - f_i^\sigma\right) + \sum_i^{\mathrm{in}} w_i c_{i\alpha} N_\alpha^\sigma \\ &= \sum_i^{\mathrm{out}} \left(f_i^{*,\sigma} - f_i^\sigma\right) - \rho^\sigma u_n + \frac{1}{2} c_s^2 N_n^\sigma, \end{aligned} \tag{17}$$

where outgoing populations are defined by $\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}}_w < 0$. Furthermore, the index n indicates the component normal to the boundary. For a set velocity boundary condition, the densities can be found using the bare velocity components,

$$\begin{aligned} \rho^\sigma u_n^\sigma &= \sum_i^{\mathrm{in}} f_i^\sigma - \sum_i^{\mathrm{out}} f_i^\sigma + \frac{1}{2} F_n^\sigma \\ &= \rho^\sigma - f_0^\sigma - 2\sum_i^{\mathrm{out}} f_i^\sigma - \sum_i^{\mathrm{par}} f_i^\sigma + \frac{1}{2} F_n^\sigma \\ \rho^\sigma &= \frac{f_0^\sigma + 2\sum_i^{\mathrm{out}} f_i^\sigma + \sum_i^{\mathrm{par}} f_i^\sigma - \frac{1}{2}F_n^\sigma}{1 - u_n^\sigma}, \end{aligned} \tag{18}$$

where parallel populations are defined as $\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}}_w = 0$ with $i \neq 0$. Furthermore, we assume that $|\hat{\boldsymbol{c}}_i \cdot \hat{\boldsymbol{n}}_w| = 1$ for $i \neq 0$. The momentum corrections N_α^σ can be computed by plugging the incoming populations from Eq. (16) back into the bare velocity components,

$$\begin{aligned} \rho^\sigma u_\alpha^\sigma &= \sum_i^{\mathrm{in}} f_i^\sigma c_{i\alpha} + \sum_i^{\mathrm{out}} f_i^\sigma c_{i\alpha} + \sum_i^{\mathrm{par}} f_i^\sigma c_{i\alpha} + \frac{1}{2} F_\alpha^\sigma \\ &= \sum_i^{\mathrm{in}} \left(2 w_i \rho^\sigma \frac{\boldsymbol{u}\cdot\boldsymbol{c}_i}{c_s^2} - w_i c_{i\beta} N_\beta^\sigma\right) c_{i\alpha} \\ &\quad + \sum_i^{\mathrm{par}} f_i^\sigma c_{i\alpha} + \frac{1}{2} F_\alpha^\sigma. \end{aligned} \tag{19}$$

The normal and parallel components of the momentum corrections can be found,

$$N_\alpha^\sigma = \frac{2}{c_s^2 A_\alpha}\left[\sum_j^{\mathrm{par}} c_{j\alpha} f_j^\sigma + A_\alpha \rho^\sigma u_\alpha - \rho^\sigma u_\alpha^\sigma + \frac{1}{2} F_\alpha^\sigma\right], \tag{20}$$

with $A_\alpha = \frac{2}{c_s^2}\sum_i^{\mathrm{in}} w_i c_{i\alpha}^2$. For the D2Q9, D3Q19 and D3Q27 lattices we have $A_\alpha = \delta_{\alpha n} + \frac{1}{3}(1 - \delta_{\alpha n})$. To correct for the mass loss given in Eq. (17), we add this back to the zero-velocity component before computing the density,

$$\begin{aligned} f_0^\sigma &= f_0^{*,\sigma} + \Delta m^\sigma + \rho^\sigma u_n \\ &= f_0^{*,\sigma} + \sum_i^{\mathrm{out}} \left(f_i^{*,\sigma} - f_i^\sigma\right) + \frac{1}{2} F_n^\sigma. \end{aligned} \tag{21}$$

Note that we have a mass loss given by $-\rho^\sigma u_n$. As this is the expected mass flux at the boundary, we do not correct for it. This simplifies the computation of the densities,

$$\rho_w^\sigma = \frac{f_0^{*,\sigma} + \sum_i^{\mathrm{out}} \left(f_i^{*,\sigma} + f_i^\sigma\right) + \sum_i^{\mathrm{par}} f_i^\sigma}{1 - u_n^\sigma}. \tag{22}$$

A. Corner Nodes

Corners between two perpendicular NEBB boundaries have to be handled separately. It is not possible to implement the corner node between two velocity boundaries without extrapolating the density. The corner node between a velocity and density boundary can be implemented. Both the incoming and buried populations are now governed by Eq. (16). The incoming populations are given by $(\hat{\boldsymbol{n}}_1 + \hat{\boldsymbol{n}}_2) \cdot \boldsymbol{c}_i > 0$ and the buried populations are given by $(\hat{\boldsymbol{n}}_1 + \hat{\boldsymbol{n}}_2) \cdot \boldsymbol{c}_i = 0$, where the normal vectors of the two intersecting walls are given by $\hat{\boldsymbol{n}}_1$ and $\hat{\boldsymbol{n}}_2$. The following equations are obtained after plugging Eq. (16) into the bare velocity components again,

$$\begin{aligned} \rho^\sigma u_\alpha^\sigma &= \sum_i^{\mathrm{in}} f_i^\sigma c_{i\alpha} + \sum_i^{\mathrm{out}} f_i^\sigma c_{i\alpha} + \sum_i^{\mathrm{buried}} f_i^\sigma c_{i\alpha} + \frac{1}{2} F_\alpha^\sigma \\ &= \rho^\sigma u_\alpha - \frac{1}{2} c_s^2 N_\alpha^\sigma + \frac{1}{2} F_\alpha^\sigma \end{aligned} \tag{23}$$

The momentum corrections are given as follows,

$$N_\alpha^\sigma = \frac{2}{c_s^2}\left[\rho^\sigma\left(u_\alpha - u_\alpha^\sigma\right) + \frac{1}{2}F_\alpha^\sigma\right] \tag{24}$$

The buried populations will be used to set the desired wall density,

$$\begin{aligned}\rho^{\text{buried},\sigma} &= \rho^\sigma - \left(f_0^\sigma + \sum_i^{\text{in}} f_i^\sigma + \sum_i^{\text{out}} f_i^\sigma\right),\\ f_i^\sigma &= \frac{w_i}{\sum_j^{\text{buried}} w_j}\rho^{\text{buried},\sigma} + w_i\rho^\sigma\frac{\boldsymbol{u}\cdot\boldsymbol{c}_i}{c_s^2}\\ &\quad - \frac{1}{2}w_i c_{i\alpha}N_\alpha^\sigma.\end{aligned} \tag{25}$$

IV. PRESCRIBED CONTACT ANGLE MODELS

To test the new multi-component boundary condition, we will perform a test case of a static droplet on a flat surface. This will test the momentum corrections taking care of external forcing on the surface and the mass corrections are examined.

A. Virtual Density Scheme for Shan-Chen

The Shan-Chen force given in Eq. (3) reaches out of the domain at the boundary nodes. The virtual densities in the solid can be computed using an averaging over the nearby fluid nodes and the contact angle is then controlled using the parameter χ^σ, [15],

$$\rho^{\text{solid},\sigma} = \chi^\sigma\frac{\sum_i^{\text{in}} w_i\rho^\sigma\left(\boldsymbol{x} + \boldsymbol{c}G_i\right)}{\sum_i^{\text{in}} w_i}, \tag{26}$$

where the incoming directions are defined by $\boldsymbol{c}_i\cdot\hat{\boldsymbol{n}}_w > 0$, with the wall normal vector pointing inside the domain. The wetting parameter is given as $\chi^\sigma = 1-\xi$ and $\chi^{\tilde{\sigma}} = 1+\xi$, where σ is the fluid making up the droplet. The surface will be hydrophilic for negative ξ and hydrophobic for positive ξ.

B. Geometric Schemes for Color-Gradient

Like for the Shan-Chen method, the computation of the Color-Gradient in Eq. (14) reaches outside of the domain. The densities inside the solid are again averaged from the nearby fluid nodes [**?**],

$$\rho^{\text{solid},\sigma} = \frac{\sum_i^{\text{in}} w_i\rho^\sigma\left(\boldsymbol{x} + \boldsymbol{c}_i\right)}{\sum_i^{\text{in}} w_i}. \tag{27}$$

These solid densities are used to compute a preliminary Color Gradient $\boldsymbol{G}^*$. To impose a given contact angle θ_c at the boundary, we need to change the angle of this vector with respect to the boundary. This can by either changing only the angle of the color gradient while keeping the norm of $\boldsymbol{G}$ unchanged [**?**], or by only changing the component of $\boldsymbol{G}$ normal to the wall [**?**],

$$\begin{aligned}\boldsymbol{G} &= |\boldsymbol{G}^*|\hat{\boldsymbol{n}}(\theta_c),\\ G_{\text{norm}} &= -\tan\left(\theta_c - \frac{1}{2}\phi\right)\sqrt{\sum_\alpha G^2_{\text{par},\alpha}}.\end{aligned} \tag{28}$$

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